

HOME AUTOMATION

Using Apple HomeKit And ESP8266

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It is an IoT based project through which you can control four home appliances using Apple HomeKit, Hey Siri app, and ESP8266 Wi-Fi module. The author's prototype is shown in Fig. 1.

A few months ago we were thinking of building a home automation system. After searching on the internet we found that such projects were mainly based on Google Assistant and Alexa. From there we got the idea to make a home automation system that could be controlled using Apple's assistant, the Hey Siri feature.

We searched this topic but most of the articles written required Apple's laptop and Homebridge app. We wanted to make the project without using these tools. Then we came across an amazing library that could make it possible, though it had an example to control only one relay. Here is the home automation project through which you can control four appliances through four relays.

Following hardware is required to make the project:

- 5V relay module
- NodeMCU board
- Jumper wires
- Bulb holder (for 230V AC)
- 60W/100W bulb
- iPhone SE or any iPhone that supports Apple HomeKit application

Circuit

Circuit connections are very simple. The Table shows connections between relay module and NodeMCU. The complete connection diagram for controlling a bulb through a relay module is shown in Fig. 6. You

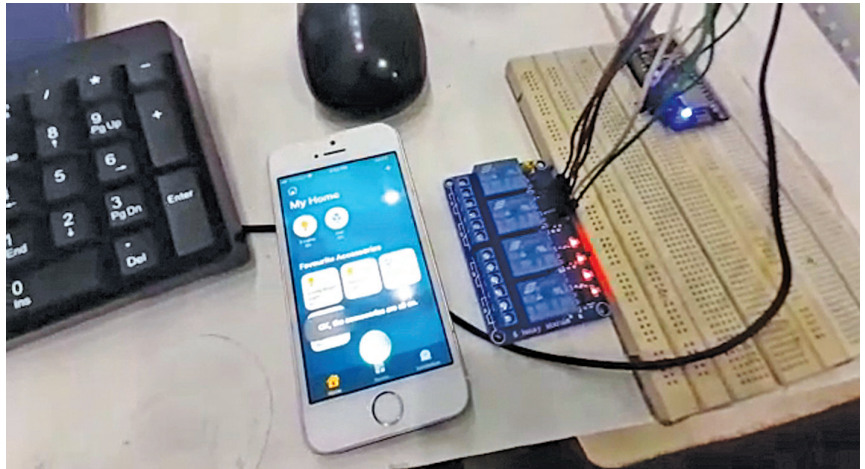


Fig. 1: Author's prototype



Fig. 2: Relay module



Fig. 4: Bulb and holder



Fig. 3: NodeMCU board

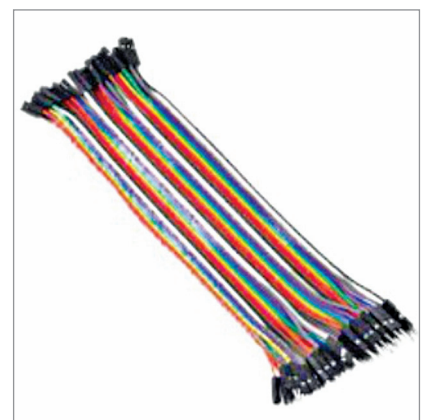


Fig. 5: Jumper wires

CIRCUIT CONNECTIONS

Pin on Relay	To NodeMCU
IN	D4
GND	G
VCC	VIN